LOCUS: Verifying OCR by Local Optimality over Counterfactual String Graphs

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Abstract

Optical character recognition engines always return a transcription, so every pipeline must decide, word by word, whether to trust it. Existing verifiers judge the claim in isolation, by engine confidence, by agreement with a second recogniser, or by asking a large vision-language model. We instead ask whether the claim is the best explanation of the image among the strings it could have been confused with. LOCUS builds around each claim a small graph of plausible single-character edits and independent readings, scores every candidate string by how much better it explains the real image than a clean rendering of itself, cancelling the recogniser's language prior, and tests whether the claim is locally optimal with a relation-aware graph neural network. The network is pretrained without human labels from engine agreement and synthetic corruptions and adapts from a few labelled words. We introduce a benchmark of real recognition errors from two engines on five public datasets covering forms, receipts, handwriting and scene text. With 25 labelled words per class, LOCUS ranks correct above incorrect transcriptions better than engine confidence, learned confidence, engine voting, Florence-2 re-recognition and a seven-billion-parameter vision-language judge on every dataset, and attains the best average macro-F1. It is significantly better than re-recognition with a strong recogniser where that recogniser is fallible, comparable where it is in-domain, and behind it on its own scene-text benchmark. Significance, recogniser, rendering and message-passing analyses and a manual audit show that the gain comes from explicit, prior-cancelled counterfactual evidence and label-free pretraining.

Keywords: OCR verification; counterfactual reasoning; graph neural networks; few-shot learning; render normalisation; document analysis

1. Introduction

Optical character recognition (OCR) engines always return a transcription, and modern end-to-end and vision–language readers return fluent text even when they are wrong [1–3].

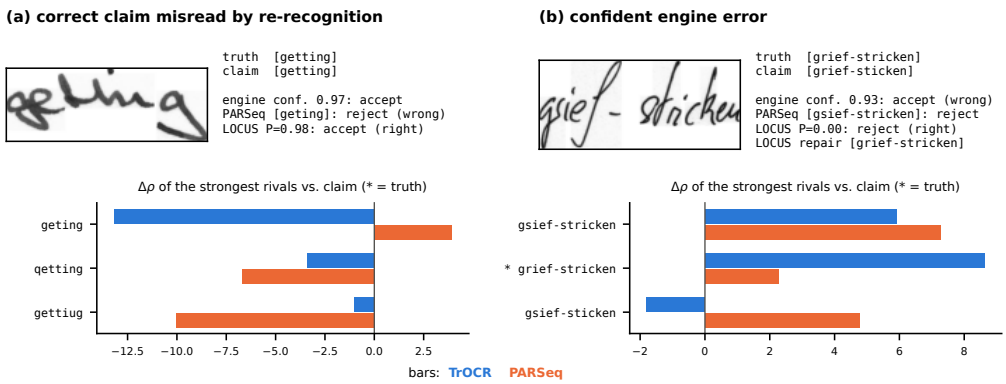


Figure 1. Why compare hypotheses instead of scoring one. Two real IAM test words. (a) A correct claim that PARSeq misreads, so re-recognition rejects it; PARSeq prefers its own reading (orange bar) but TrOCR strongly disfavours it (blue bar), and LOCUS, which weighs both verifiers' evidence for every rival, accepts the claim (this happens for 312 of the 419 correct IAM claims that PARSeq misreads). (b) A confident engine error that engine confidence accepts; LOCUS rejects it and proposes the correct string. Bars: render-normalised evidence of the strongest rivals relative to the claim ($*$: ground truth).

Every consumer of OCR (key-information extraction from receipts and forms, archival search, accessibility, data entry) must therefore decide, word by word, whether a transcription can be trusted or must be routed to a human. This is *OCR verification*: given a word image x and a claim h produced by an arbitrary, possibly black-box engine, predict whether h is exactly what is written in x. The three families of verifiers used in practice each have a characteristic blind spot (Figure 1). **Engine confidence** is unavailable for many engines and overconfident on fluent, plausible errors. **Re-recognition and voting** (accept h iff a second recogniser reads the same string [4–6]) is a binary test that inherits every bias of the second recogniser. **Vision–language judges** [7, 8] are billions of parameters large, cannot be adapted from a handful of labels and, as we show, are poor at the character-exact judgements verification needs.

Key idea. The right question is not "does the image look like h?" but "is h the best explanation of the image among the strings it could have been confused with?" An OCR error is a near miss: a confusable substitution (`O`/`0`, `l`/`1`), a dropped or inserted glyph, or a case flip. We build around every claim a small *counterfactual string graph* of its OCR-plausible edits and test whether the claim is a *local optimum* of visual evidence on it. Two problems must be solved. First, the score must be *visual*: a recogniser's likelihood $s(v \mid x)$ mixes how well v explains the pixels with how plausible v is as a string, so rare but correct strings are penalised. We cancel this prior with a *render-normalised likelihood ratio* $\rho(v) = s(v \mid x) - s(v \mid R(v))$, where $R(v)$ is a clean rendering of v. Second, the comparison must be learned from very few labels: a relation-aware graph network with an explicit local-optimality readout is pretrained *without human labels* on images where two engines agree, paired with synthetic corruptions whose repair is known.

Contributions. (1) A formulation of OCR verification as testing the local optimality of a claim among OCR-plausible counterfactuals, which also yields a repair proposal (Section 3). (2) Render-normalised evidence that removes the recogniser's string prior, computed with

two frozen, architecturally different recognisers. (3) LOCUS, a relation-aware GNN over counterfactual string graphs, pretrained label-free and adapted from $k \in \{5, \ldots, 100\}$ labelled words per class. (4) A verification benchmark of *real* OCR errors on five public datasets (forms, camera and scanned receipts, handwriting, scene text) with two engines, and a comparison with engine and learned confidence, voting, three re-recognition baselines and a 7B vision–language judge (Sections 4–5). (5) An honest analysis: review cost, transfer, repair, document-level significance, recogniser dependence, rendering sensitivity, message passing, and a manual audit with representative qualitative examples (Sections 6–7).

2. Related Work

OCR confidence and failure prediction. Engine confidence is the default verifier, and learned failure predictors regress the true-class probability from a classifier's own features [9]. Both score a single hypothesis in isolation and need access to engine internals; neither compares the claim with its visual competitors.

Post-OCR correction with large models. Since 2024 post-OCR processing has moved to large (multimodal) language models. Multimodal LLMs can transcribe and post-correct historical documents better than conventional pipelines [10], but LLM post-correction gives "no free lunches": gains depend on language, model size and input error rate, and correct text can be damaged [11]; the ICDAR 2026 HIPE-OCRepair competition makes LLM-assisted post-correction a shared task [12]. Vision–language models also act as judges or repair agents for document parsing: DOCR-Inspector fine-tunes a 7B model to enumerate parsing errors [8], and OCR-EDR trains a diagnosis-and-repair agent that may request renderings of its edits [13]. These systems target page-level parsing, need large curated training sets and are orders of magnitude more expensive per word; we include a 7B judge [7] as a baseline.

End-to-end OCR models and their hallucinations. Unified models such as GOT-OCR2.0 [14] and VLM-based PDF linearisers such as olmOCR [15] read whole pages, and OCRBench v2 shows that even the strongest multimodal models still make frequent fine-grained recognition errors [1]. Their errors are often *plausible*: multimodal LLMs hallucinate text on degraded images [2] and a low character error rate can hide hallucinated content [3]. This motivates an engine-agnostic verifier that asks whether the image supports *this* string better than its near-miss rivals. Strong recognisers such as TrOCR [5] and PARSeq [4] are natural, frozen verifiers in this setting.

Contrast against a reference. Visual contrastive decoding reduces hallucination by contrasting a model's outputs on the real and on a distorted image [16]. Our render-normalised ratio applies the same principle with a reference *specific to each hypothesis*: a clean rendering of the very string being tested.

Graph networks for OCR verification. Character-level graphs comparing real and synthetic glyphs have been used to verify Hindi OCR [17], and Saxena *et al.* [18] represent each word as a relational line graph of real-versus-rendered character comparisons, verified few-shot by a GNN on a single private low-resolution dataset. Both require character segmentation of the word image and compare characters, not alternative readings. LOCUS needs no segmentation: its nodes are *hypotheses* and its edges are edit operations between them, encoded with relation-aware attention [19].

3. Proposed Method: LOCUS

Problem statement. An OCR engine $\mathscr{E}$ maps a word (or text-line) image x to a claim $h = \mathscr{E}(x)$. The engine is a black box: we observe only h. With t the true transcription, verification predicts $y = \mathbf{1}[h = t]$ (CORRECT vs. INCORRECT) and, optionally, proposes a repair $\hat{t}$. We study the label-efficient regime with only $k \in \{5, 10, 25, 50, 100\}$ labelled words per class plus an unlabelled pool of images with engine outputs, and also report full supervision. Figure 2 gives an overview on a real test word and Algorithm 1 summarises inference.

3.1 Counterfactual string graphs

For each claim h we build a graph $G(h) = (\mathscr{V}, \mathscr{E})$ whose nodes are strings: the claim and up to $M = 28$ counterfactual neighbours generated without any ground truth, in this priority order: (i) the greedy readings g_1, g_2 of two frozen verifier recognisers V_1, V_2; (ii) *reading-aligned edits*, the single substitutions, insertions and deletions that move h one step towards g_1 or g_2 under a Levenshtein alignment; (iii) *confusion edits*: substitutions within visually confusable groups (e.g. {O,0,o,Q,D}, {I,l,1,|}, {5,S}), case flips and single deletions. Every neighbour is linked to the claim by an edge typed by its edit operation (seven types); *sibling* edges link neighbours that edit the same position (competing explanations of one glyph) and *path* edges link each reading to the single edits leading to it, giving a one-hot edge type $e_{uv} \in \{0,1\}^{11}$. OCR errors are near misses, so the truth lies inside $G(h)$ for most real errors (Table 2), which makes repair possible.

3.2 Render-normalised evidence

Let $s_V(v \mid x) = \sum_i \log p_V(v_i \mid v_{<i}, x)$ be the teacher-forced log-likelihood of string v under a frozen autoregressive recogniser V. It conflates visual fit with the prior that V's decoder has learned over strings: a correct but rare string (a serial number, a surname, lower-case text under a recogniser trained on upper-case receipts) scores low. With $R(v)$ a clean rendering of v in a standard font at the height of x, we define

$$\rho_V(v) \;=\; s_V(v \mid x) \;-\; s_V\big(v \mid R(v)\big). \tag{1}$$

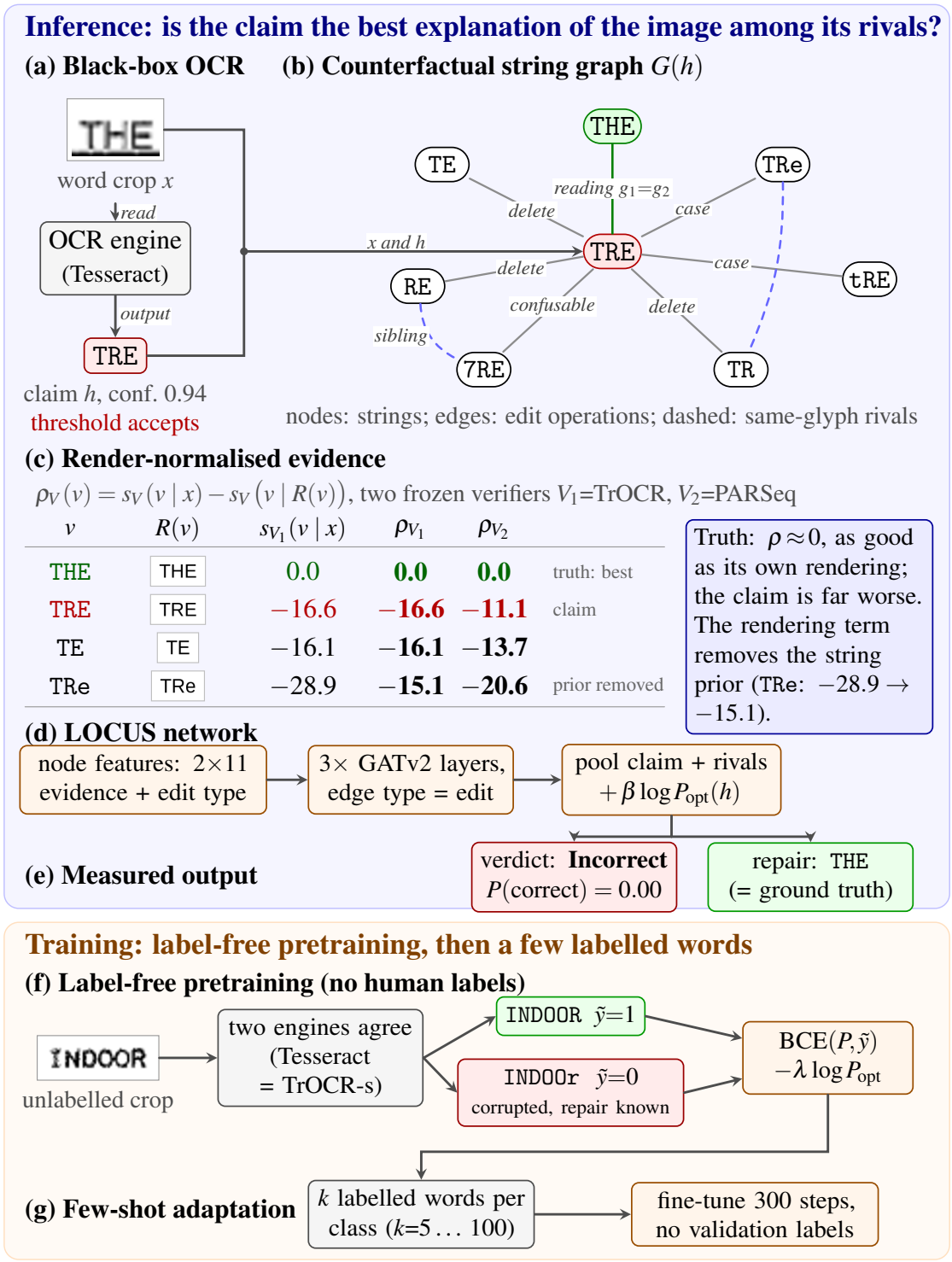
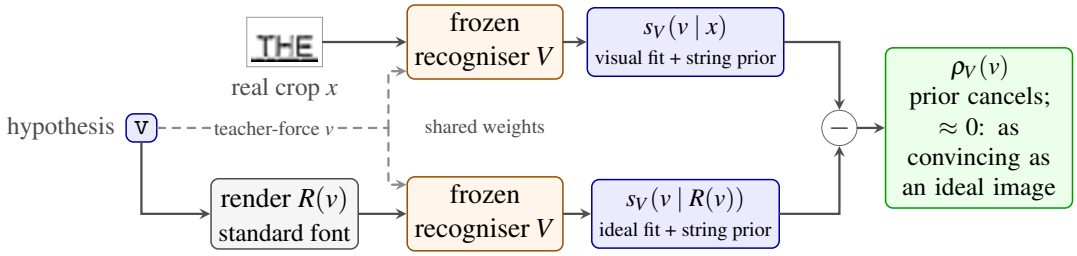


Figure 2. LOCUS at a glance, on a real FUNSD test word (truth THE). (a) A black-box engine reads TRE. (b) LOCUS builds the claim's counterfactual string graph. (c) Every hypothesis is scored by its render-normalised likelihood ratio under two frozen recognisers. (d) A relation-aware graph network compares the claim with its rivals and (e) rejects it, proposing the ground truth as repair. (f) Pretraining uses no human labels; (g) adaptation uses k labelled words per class. All strings, images and numbers shown are real.

(a) The ratio for one hypothesis v



(b) Measured on the real word (truth `THE`; TrOCR as V)

v	$R(v)$	$s_V(v \mid x)$	$s_V(v \mid R(v))$	$\rho_V(v)$
THE	THE	0.0	0.0	**0.0**
TRE	TRE	−16.6	0.0	**−16.6**
TRe	TRe	−28.9	−13.8	**−15.1**

`TRE` and `TRe` differ only in case. Raw scores differ by 12.3, mostly string prior (`TRe` scores −13.8 on its own clean rendering); after normalisation they differ by 1.5.

Figure 3. Render normalisation. (a) The same frozen recogniser scores hypothesis v twice, teacher-forced on the real crop and on a clean rendering of v; the difference cancels the decoder's prior over strings. (b) Real TrOCR values for the word of Figure 2: the case variant `TRe` looks far less likely than `TRE` on the raw score only because TrOCR's decoder dislikes the string, which the ratio removes.

Under the factorisation $\log p_V(v \mid x) \approx \log p_{\text{vis}}(x \mid v) + \log \pi_V(v) - \log Z(x)$ the string prior $\pi_V(v)$ appears in both terms and cancels; $\rho_V(v) \approx 0$ means "x is as convincing a picture of v as a perfect rendering", while strongly negative values flag visual disagreement (Figure 3). We also record the token-level minima of the log-likelihood and of the ratio (an error is often a single character) and the cosine distance between V's decoder states on x and on $R(v)$. Every quantity is computed with two recognisers of different design: V_1=TrOCR-base [5] (ViT encoder, BPE decoder; the printed, handwritten or scene-text checkpoint matching the domain) and V_2=PARSeq [4] (character-level permuted autoregressive decoder). Neither is fine-tuned. Each node thus carries 2×11 evidence features, seven claim-relative differences $\phi(v) - \phi(h)$, its edit type and its relative edit position.

3.3 The LOCUS network

Node features are standardised with statistics of the *unlabelled* pool and embedded linearly. Three relation-aware GATv2 layers [19] with edge features and residual LayerNorm update the node states:

$$h_u^{(\ell+1)} = \mathrm{LN}\Big(h_u^{(\ell)} + \sigma\Big(\textstyle\sum_{v \in \mathcal{N}(u)} \alpha_{uv}^{(\ell)} W^{(\ell)} h_v^{(\ell)}\Big)\Big),$$
$$\alpha_{uv}^{(\ell)} = \mathrm{softmax}_v\Big(a^\top \mathrm{LReLU}\big(W[h_u \| h_v \| e_{uv}]\big)\Big). \tag{2}$$

Local-optimality readout. A plausibility score $q_v = w_q^\top \mathrm{MLP}(h_v)$ defines $P_{\text{opt}}(v \mid G) = \exp q_v / \sum_{u \in \mathcal{V}} \exp q_u$, the probability that v is the best explanation of x within its neighbourhood. The verdict combines the claim's state, an attention pool and a max pool over its rivals, and the

Input: word image x, engine claim h; frozen recognisers V_1, V_2; renderer R; trained LOCUS network.
Output: $P(\text{CORRECT} \mid x, h)$ and a repair proposal $\hat{t}$.

1. $g_1 \leftarrow V_1(x)$, $g_2 \leftarrow V_2(x)$ *(greedy readings)*
2. $\mathscr{V} \leftarrow \{h, g_1, g_2\} \cup \text{AlignedEdits}(h, g_1, g_2) \cup \text{ConfusionEdits}(h)$, truncated to $M{+}1$ nodes
3. **for** each $v \in \mathscr{V}$ and $V \in \{V_1, V_2\}$: $\rho_V(v) \leftarrow s_V(v \mid x) - s_V(v \mid R(v))$, plus token minima and latent distance
4. build typed edges (edit, sibling, path) and node features $[\phi(v) \,\|\, \phi(v) - \phi(h) \,\|\, \text{edit type, position}]$
5. run 3 GATv2 layers; compute $P_{\text{opt}}(\cdot \mid G)$ and $\text{logit}\, P(y = 1 \mid x, h)$
6. average with the support-fitted logistic head on $[z \,\|\, \log P_{\text{opt}} \,\|\, \text{logit}]$ and character evidence
7. **return** P, and $\hat{t} = \arg\max_{v \neq h} P_{\text{opt}}(v \mid G)$ if $P < 0.5$

Algorithm 1. LOCUS inference for one claim.

log-probability that the claim itself is the local optimum,

$$\text{logit}\, P(y = 1 \mid x, h) = \text{MLP}\big([h_h \,\|\, \textstyle\sum_v a_v h_v \,\|\, \max_v h_v]\big) + \beta \log P_{\text{opt}}(h \mid G), \tag{3}$$

and the same distribution gives the repair $\hat{t} = \arg\max_{v \neq h} P_{\text{opt}}(v \mid G)$. Complementing the string-level graph, *character evidence* summarises per-position counterfactual margins of the claim (drop in s_{V_1} under confusable replacement, deletion or case flip, token entropy and token-level ratio), pooled by mean/min/max.

3.4 Label-free pretraining and few-shot adaptation

On the unlabelled training images we keep those on which two OCR engines agree (a high-precision proxy for correctness) and pair each with (i) the agreed string as a pseudo-correct claim and (ii) an OCR-style corruption of it (confusable substitution, case flip, deletion or insertion) as a pseudo-incorrect claim whose repair is the original. LOCUS is trained on these graphs with $\mathscr{L} = \text{BCE}\big(P(y = 1 \mid x, h), \tilde{y}\big) - \lambda \log P_{\text{opt}}(\tilde{t} \mid G)$ ($\lambda = 0.5$), where $\tilde{y}$ is the pseudo-label and $\tilde{t}$ the node holding the pseudo-truth; no human label is used. Given k labelled words per class we (a) fine-tune the pretrained network for 300 steps (AdamW, lr $3 \cdot 10^{-4}$) and (b) fit an ℓ_2-regularised logistic regression on the frozen embedding $[z \,\|\, \log P_{\text{opt}} \,\|\, \text{logit}]$ and the character evidence, with regularisation chosen by cross-validation *on the support set only*; the verdict averages (a) and (b). No validation labels are used in the few-shot regime.

4. Experimental Setup

4.1 A verification benchmark with real OCR errors

Public OCR datasets provide images and transcriptions but no verification labels. We create them by running real OCR engines and labelling each output against the ground truth (Figure 5a), so every INCORRECT example is a genuine engine error, not a synthetic perturbation. Five

Table 1. Benchmark. Five public datasets with real OCR errors. Error rate: share of engine outputs that differ from the ground truth (exact match; SROIE case-insensitive; IIIT5K case-insensitive alphanumeric).

Dataset	domain	unit	#train	#test	avg. len.	engine	err. train (%)	err. test (%)
FUNSD	scanned forms	word	8,000	4,000	5.1	Tesseract 5	28.7	36.9
CORD	camera receipts	word	8,000	2,356	4.9	Tesseract 5	37.0	39.9
SROIE	scanned receipts	line	8,000	4,000	11.5	Tesseract 5	47.8	47.0
IAM	handwriting	word	8,000	4,000	4.2	TrOCR-small	48.4	46.4
IIIT5K	scene text	word	2,000	2,999	5.0	Tesseract 5	14.2	29.0

Table 2. Counterfactual graphs on the test sets: average hypotheses per claim and share of real engine errors whose ground truth equals TrOCR's reading, PARSeq's reading, or *any* node of $G(h)$.

Dataset	nodes	truth $= g_1$ (%)	truth $= g_2$ (%)	truth $\in G(h)$ (%)
FUNSD	17.2	31.3	72.4	82.8
CORD	15.3	70.4	89.5	94.9
SROIE	21.2	89.5	44.1	92.6
IAM	17.2	14.4	46.5	55.4
IIIT5K	15.9	86.8	95.3	97.0

public datasets cover four domains (Table 1, Figure 4): **FUNSD** [20], word boxes in noisy scanned forms (the Hugging Face release stores boxes on the LayoutLM 0–1000 grid; they must be rescaled to pixels before cropping); **CORD** [21], words in camera-captured receipts (official train and validation splits form the training pool); **SROIE** [22], text lines of scanned receipts, whose ground truth is entirely upper-case, so comparison is case-insensitive; **IAM** [23], handwritten English words; and **IIIT5K** [24], scene-text words, for which we follow the standard case-insensitive alphanumeric protocol and give *every* verifier, including LOCUS and the VLM judge, the claim in this canonical form. Official splits are kept and pools are capped at 8,000 training and 4,000 test crops (fixed seed). The main engine is Tesseract 5 [25] (LSTM, `psm 7`); on handwriting Tesseract is degenerate (93% word error), so IAM verifies TrOCR-small-handwritten. A second engine, TrOCR-small, is used for engine transfer. Test error rates span 29–47%.

What counts as an error. Labels inherit the datasets' conventions and noise. A share of incorrect claims differ from the ground truth only in case or punctuation (mostly fragments of neighbouring glyphs inside the word box), and our manual audit finds annotation errors and clipped crops among them (Section 7). We keep these claims (an engine that emits `Barbara |` for the field `Barbara` is wrong for a downstream consumer) but also report every method on genuine character errors only (Section 6.5). Table 2 shows that the counterfactual graph contains the truth for 83–97% of real errors on printed and scene text and for 55% on handwriting, where many errors span several characters.

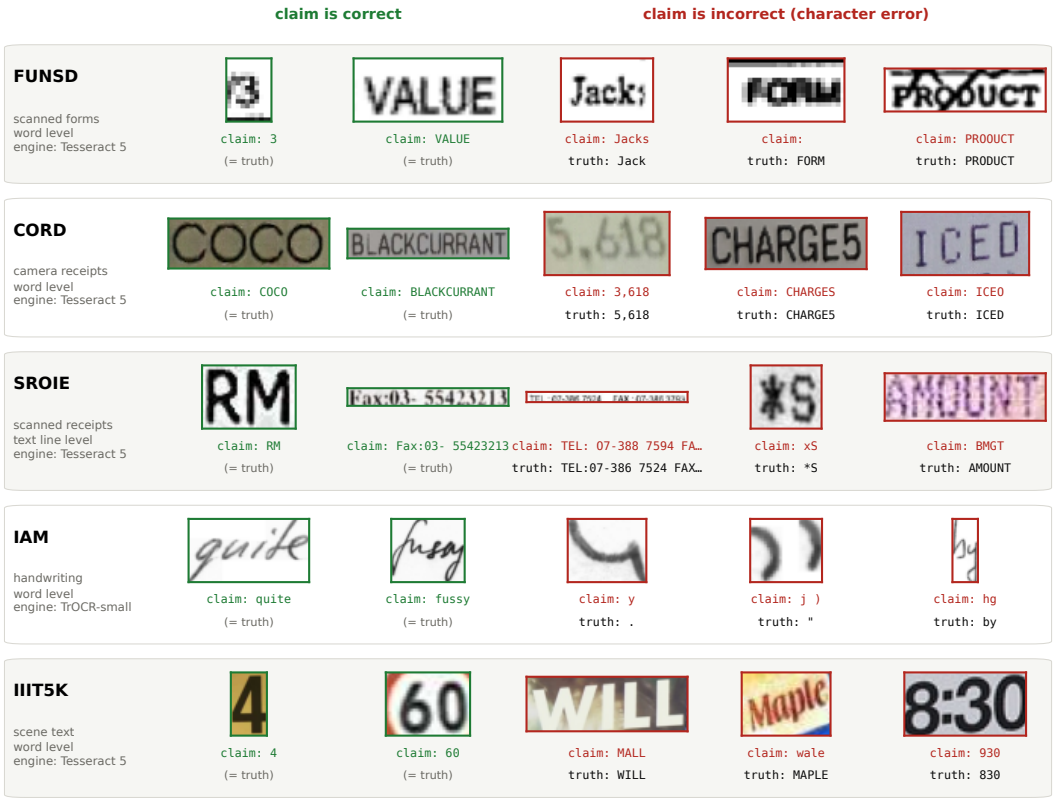


Figure 4. The five verification benchmarks. Real test crops with the claim of the main OCR engine, drawn by a seeded random choice (two correct claims and three character errors per dataset; none chosen by hand). Errors range from single confusable glyphs (`CHARGES`/`CHARGE5`, `3,618`/`5,618`) to empty or noise outputs.

4.2 Baselines, protocol and metrics

All baselines see the same test words, and every threshold is fitted on the same k-shot support set as LOCUS (maximising macro-F1). **Engine confidence**: Tesseract's mean word confidence or TrOCR-small's mean token probability (temperature scaling is monotone and leaves ranking metrics unchanged). **ConfidNet-style learned confidence** [9]: an MLP trained on the support set from the engine confidence and the verifier's likelihood statistics of the claim. **Re-recognition agreement**: accept iff an independent recogniser reads the same string: TrOCR-base [5], PARSeq [4] and Florence-2 [6]; for PARSeq we also report a *graded* variant ranked by its sequence confidence (agree $\rightarrow +c$, disagree $\rightarrow -c$), the strongest re-recognition baseline. **Two-engine voting**: accept iff Tesseract and TrOCR-small agree. **VLM judge**: Qwen2.5-VL-7B-Instruct [7] asked whether the image shows exactly the claimed text, scored by $P(\texttt{yes})/(P(\texttt{yes})+P(\texttt{no}))$ at the natural 0.5 threshold and at an *oracle* threshold tuned on the test set. **Strong learned baselines on our evidence**: logistic regression and gradient-boosted trees on flattened LOCUS evidence and/or character evidence, DeepSets over rivals (no edges), and LOCUS from scratch.

For each k we draw five class-balanced support sets (seeds 0–4); the test set keeps its natural

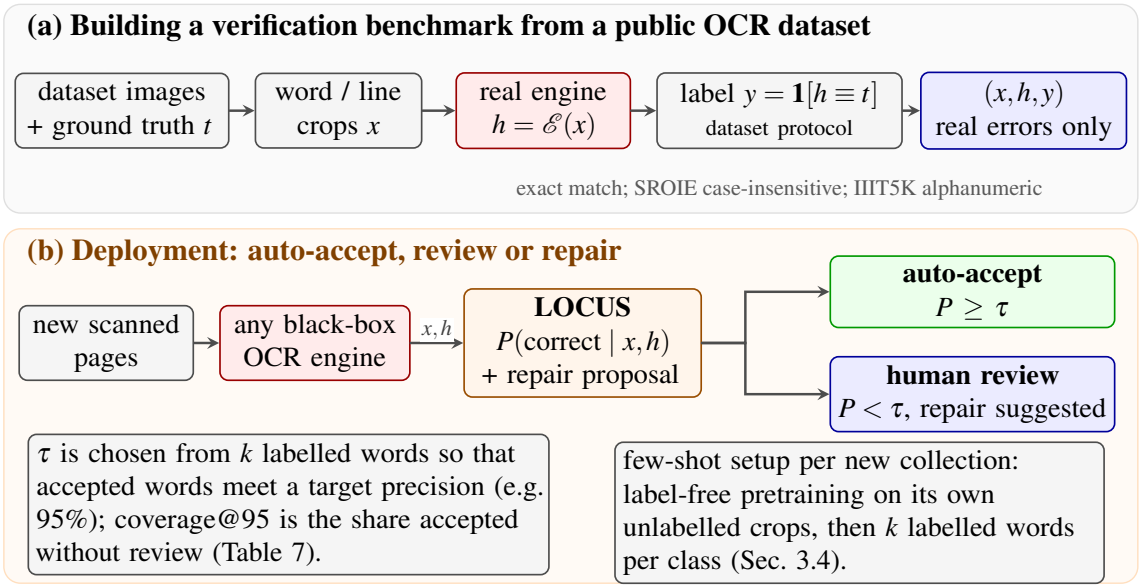


Figure 5. From benchmark to deployment. (a) Each benchmark labels real engine outputs against the dataset's own ground truth under its protocol. (b) In a digitisation pipeline, words above a precision-targeted threshold are accepted automatically and the rest go to a reviewer with a repair suggestion.

class ratio, and we report the ensemble of the five models. Metrics: **macro-F1** (primary), **AUROC**, **AURC** (area under the risk–coverage curve, lower is better), **coverage@95** (share of words that can be accepted while the accepted set stays $\geq 95\%$ correct) and top-1 **repair accuracy**. In every column of a comparison table the best entry is **bold on green** and the second best underlined on blue. **Implementation:** hidden size 64, 3 GATv2 layers with 4 heads, dropout 0.2; pretraining 15 epochs (AdamW, lr 10^{-3}, batch 64); rendering in Liberation Sans; the trainable GNN has 0.05 M parameters and evidence extraction runs at ≈18 words/s on one RTX A6000.

5. Results and Discussion

5.1 Main comparison

Tables 3 and 4 compare LOCUS with all baselines at $k = 25$ (other k and all metrics: Supplementary Section S6).

General-purpose verifiers are far from sufficient. Engine confidence, its learned refinement, engine voting, Florence-2 re-recognition and the 7B VLM judge all trail LOCUS by large margins on every dataset. Despite 8.3B parameters the VLM judge reaches only 0.760 macro-F1 on FUNSD and 0.700 on IAM, and an oracle threshold barely helps: its errors are errors of *ranking*, not of calibration. Character-exact judgements (`O` vs. `0`, a missing comma, a case flip) are what instruction-tuned VLMs gloss over.

Re-recognition with a strong recogniser is the baseline to beat. Accepting a claim iff

Table 3. Main comparison, macro-F1 at $k = 25$ labelled words per class (learned methods: ensemble of five seeds; threshold baselines: mean over the five supports). FUNSD and CORD were used for development; SROIE, IAM and IIIT5K are held out. †Threshold tuned on the test set (upper bound); ‡the best baseline chosen separately for every dataset in hindsight (both excluded from the colour marks).

Method	FUNSD	CORD	SROIE	IAM	IIIT5K	Avg.
Baselines						
Engine MSP confidence	.756	.777	.780	.802	.788	.781
ConfidNet-style learned conf. [9]	.771	.842	.826	.801	.922	.832
Two-engine voting	.609	.772	.976	.759	.862	.796
Re-OCR: TrOCR-base [5]	.605	.811	**.979**	.758	.952	.821
Re-OCR: Florence-2 [6]	.647	.615	.478	.369	.823	.586
Re-OCR: PARSeq [4]	.946	.973	.742	.879	**.993**	.907
Re-OCR: PARSeq, graded	.946	.973	.742	.879	**.993**	.907
VLM judge: Qwen2.5-VL-7B [7]	.760	.891	.866	.700	.888	.821
VLM judge, oracle threshold†	.762	.902	.883	.719	.928	.839
Best baseline per dataset, chosen in hindsight‡	.946	.973	.979	.879	.993	.954
Strong learned baselines on our evidence						
LR on flattened graph evidence	**.948**	.955	.955	.904	.969	.946
LR on character evidence	.935	.971	.959	.887	.965	.943
GBDT on all evidence	.922	.961	.962	.893	.955	.939
DeepSets over rivals (no edges)	.942	.977	.968	**.909**	.965	.952
LOCUS-GNN from scratch	.933	.969	.961	.908	.962	.947
Ours						
LOCUS zero-shot (no labels)	.918	.904	.798	.825	.968	.883
LOCUS (k-shot)	.943	**.978**	.973	.906	.971	**.954**

Table 4. Main comparison, AUROC at $k = 25$ (same protocol as Table 3).

Method	FUNSD	CORD	SROIE	IAM	IIIT5K	Avg.
Baselines						
Engine MSP confidence	.861	.892	.859	.878	.938	.886
ConfidNet-style learned conf. [9]	.845	.929	.912	.861	.980	.905
Two-engine voting	.680	.807	.976	.767	.905	.827
Re-OCR: TrOCR-base [5]	.675	.841	.978	.766	.966	.845
Re-OCR: Florence-2 [6]	.715	.685	.578	.522	.879	.676
Re-OCR: PARSeq [4]	.939	.975	.763	.885	.991	.911
Re-OCR: PARSeq, graded	**.976**	.986	.760	.914	**.995**	.926
VLM judge: Qwen2.5-VL-7B [7]	.846	.945	.941	.788	.974	.899
VLM judge, oracle threshold†	.846	.945	.941	.788	.974	.899
Best baseline per dataset, chosen in hindsight‡	.976	.986	.978	.914	.995	.970
Strong learned baselines on our evidence						
LR on flattened graph evidence	.975	**.995**	.990	.970	.991	**.984**
LR on character evidence	.973	.992	.990	.959	.993	.981
GBDT on all evidence	.969	.988	.991	.964	.990	.981
DeepSets over rivals (no edges)	.971	.994	**.992**	**.972**	.990	**.984**
LOCUS-GNN from scratch	.969	.992	.990	.969	.990	.982
Ours						
LOCUS zero-shot (no labels)	.949	.977	.966	.909	.983	.957
LOCUS (k-shot)	.974	.993	**.992**	.969	.991	**.984**

PARSeq reads the same string is by far the strongest baseline, and its graded variant more so. LOCUS improves on it where PARSeq is fallible: on IAM handwriting by +0.027 macro-F1 and +0.055 AUROC, and on CORD receipts (macro-F1 0.978 vs. 0.973; AUROC gain significant, Table 9). On FUNSD the two are statistically tied, and on IIIT5K, the scene-text benchmark on which PARSeq was designed and reads almost every word correctly, PARSeq agreement remains better (0.993 vs. 0.971). On held-out SROIE the ranking of baselines flips: TrOCR's printed checkpoints were fine-tuned on SROIE receipts, so TrOCR re-recognition is strong (.979 macro-F1) while PARSeq agreement fails on these long receipt lines (.742); LOCUS is slightly

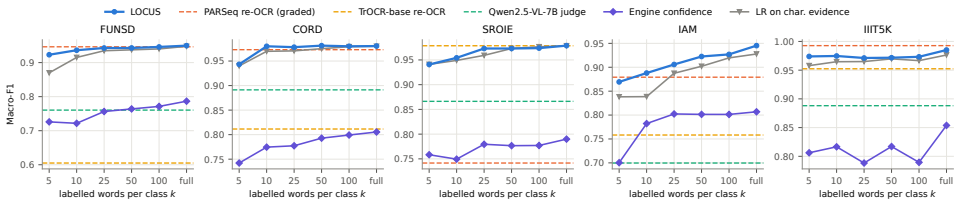


Figure 6. Label efficiency. Macro-F1 versus labelled words per class (ensembles of five seeds; dashed: label-free baselines). AUROC curves: Supplementary Figure S2.

below the former in macro-F1 (.973) and above it in AUROC (.992 vs. .978, Table 9). No single re-recognition baseline is the best choice on every dataset (PARSeq on forms and scene text, TrOCR on SROIE, neither on handwriting), whereas LOCUS is within 0.006 macro-F1 of the best baseline or better on four of the five datasets. Its average macro-F1 (.954) is the highest of all methods in Table 3: well above the best single baseline (.907, PARSeq), equal to choosing the best baseline separately for every dataset in hindsight (.954), and marginally above simple learners trained on our own counterfactual evidence (.939–.952).

Learned baselines on our evidence and the label-free model. Logistic regression, gradient boosting and DeepSets on the *same* counterfactual evidence are competitive with LOCUS (middle blocks): most of the gain over general-purpose baselines comes from the counterfactual, render-normalised *evidence* itself, and Section 6.3 shows that message passing adds little on top of it. Pretrained only on engine agreement and synthetic corruptions, zero-shot LOCUS already exceeds engine confidence, learned confidence, Florence-2 and the VLM judge in AUROC on every dataset, and engine voting on four of five (all except SROIE, where both voting engines include a SROIE-trained recogniser).

5.2 Label efficiency and render normalisation

Figure 6 shows that LOCUS is close to its fully supervised accuracy with $k = 10$–25 labelled words per class on the document datasets, and keeps improving with more labels on IAM; seed variance shrinks rapidly with k (Supplementary Section S6). Supplementary Figure S7 isolates Equation (1) on a single score: the raw TrOCR likelihood of the claim is a poor detector wherever the recogniser's string prior is mismatched with the domain, and subtracting the likelihood of the claim's own rendering repairs it. In the full model, where a second verifier and agreement features overlap, removing render normalisation costs little (Table 5); with TrOCR as the only verifier it helps where TrOCR's prior is mismatched (SROIE .956 vs. .902, CORD .973 vs. .967, IIIT5K .957 vs. .951), is neutral on FUNSD and slightly harmful on IAM, and with PARSeq alone it makes no difference (Supplementary Table S15): render normalisation corrects a *biased* recogniser and is neutral for an unbiased one.

Table 5. Ablations at $k = 25$, macro-F1 (ensemble of five seeds). Δ: mean change w.r.t. the full model over the datasets where the variant was run (AUROC: Supplementary Table S6).

Variant	FUNSD	CORD	SROIE	IAM	IIIT5K	Δ
LOCUS (full)	.939	.979	**.974**	.902	.970	
+ CV-selected prior/anchored head	.944	.961	.965	.903	.972	-0.004
+ linear probe on embedding	.943	.970	.965	.901	.967	-0.004
w/o verifier-agreement node features	.937	.978	.963	.902	.958	-0.005
w/o edit-graph GNN (char evidence LR)	.935	.971	.959	.887	.965	-0.009
w/o label-free pretraining	.932	.969	.962	.908	.962	-0.006
w/o local-optimality term	.943	.980	.968	.899	.967	-0.002
w/o repair loss	.941	**.981**	**.974**	.899	.965	-0.001
w/o render-normalisation	.942	.977	.967	.893	.973	-0.002
w/o latent render distance	.940	.978	.971	**.914**	.972	+0.002
likelihood only (no render)	.940	.979	.969	.913	.975	+0.002
TrOCR verifier only	.942	.980	.971	.881	.975	-0.003
PARSeq verifier only	.938	.978	.969	.898	.970	-0.002
star edges only	.943	**.981**	.968	.898	.972	-0.001
no message passing	**.945**	.978	.969	.895	**.979**	+0.000
GCN backbone	.942	.978	.965	.905	.969	-0.001
GraphSAGE backbone	.933	.978	.970	.906	.966	-0.002
GIN backbone	.940	.972	.955	.889	.964	-0.009

Table 6. Transfer. Left: engine transfer (train on one engine's claims, test on the other's; in-engine: trained and tested on the target engine). Right: dataset transfer, macro-F1 (rows: training dataset, fully supervised; shaded: in-domain).

		$k = 25$			full		
Dataset	Train→Test	F1 in	F1 tr.	AUROC tr.	F1 in	F1 tr.	AUROC tr.
FUNSD	Tesseract→TrOCR-s	.955	.838	.963	.959	.958	.988
FUNSD	TrOCR-s→Tesseract	.941	.927	.964	.951	.947	.976
CORD	Tesseract→TrOCR-s	.956	.689	.974	.971	.891	.989
CORD	TrOCR-s→Tesseract	.980	.817	.982	.979	.879	.993
SROIE	Tesseract→TrOCR-s	.629	.745	.862	.716	.773	.884
SROIE	TrOCR-s→Tesseract	.973	.776	.922	.979	.787	.941
IIIT5K	Tesseract→TrOCR-s	.990	.980	.995	.984	.993	.996
IIIT5K	TrOCR-s→Tesseract	.971	.972	.988	.985	.966	.991

Train\Test	FUNSD	CORD	SROIE	IAM	IIIT5K
FUNSD	.949	.977	.861	.905	.938
CORD	.942	.981	.959	.882	.914
SROIE	.620	.823	.978	.779	.946
IAM	.880	.878	.734	.944	.925
IIIT5K	.946	.973	.767	.899	.978

5.3 Ablations

Table 5 reports every component. The clearest effects are label-free pretraining and operating on the edit graph at all (versus the same character evidence without rival hypotheses). Several conceptually central components are individually *redundant* in the full model: removing message passing, the local-optimality term, the repair loss or render normalisation changes macro-F1 by less than a point on most datasets, because the two verifiers and the agreement features carry overlapping information; GCN and GraphSAGE backbones perform like GATv2 and GIN is slightly worse (-0.009). The value of LOCUS lies in *what* is compared (the claim against its counterfactual rivals, under prior-cancelled evidence) more than in the network that compares them.

5.4 Transfer across engines and datasets

A verifier that must be retrained for every engine is of limited use. Trained on Tesseract claims and tested on TrOCR-small claims (and vice versa), the fully supervised *ranking* transfers with an AUROC loss of at most 0.006 on FUNSD, CORD and IIIT5K (at most 0.023 at $k = 25$), but

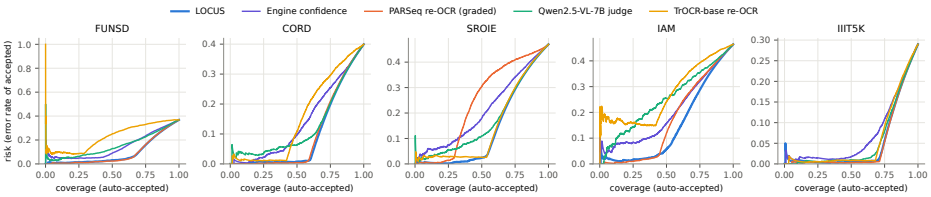


Figure 7. Risk–coverage. Error rate among auto-accepted words as a function of the fraction accepted ($k = 25$; lower is better).

SROIE loses 0.02–0.07, and the 0.5 decision threshold transfers poorly everywhere because the two engines' error rates and types differ; a new engine therefore needs at least threshold re-calibration (Table 6). The ranking transfers well across all twenty dataset pairs (AUROC .892–.995; e.g. FUNSD→CORD .995 vs. .996 in-domain, CORD→FUNSD .968 vs. .979). The weakest transfers are those *to* handwriting (SROIE→IAM .892, CORD→IAM .954) and from receipt lines to forms (SROIE→FUNSD .941). As for engines, macro-F1 at the fixed 0.5 threshold transfers much less well than the ranking (e.g. SROIE→FUNSD .620).

5.5 Practical use: human review, repair and cost

The operational question in a digitisation pipeline (Figure 5b) is how many words a human must check to guarantee the quality of the rest (Figure 7, Table 7). At 95% accepted precision, LOCUS requires reviewing 44.7% of SROIE lines (TrOCR re-recognition 45.0%, PARSeq 69.5%, the VLM judge 76.2%, engine confidence 90.9%) and 51.4% of IAM words (PARSeq 54.6%, the VLM judge 94.8%). On FUNSD, CORD and IIIT5K, LOCUS and PARSeq are within two points of each other (FUNSD 37.2% vs. 35.4%, CORD 37.4% vs. 38.6%, IIIT5K 25.6% vs. 25.3%), while engine confidence and the VLM judge require 1.6–2.3× more review than LOCUS on the three receipt and form datasets. At the strictest target (98%) PARSeq is better on FUNSD and IAM. For repair (Table 8), the truth lies in the counterfactual graph for most errors, but substituting the better verifier's reading is at least as good a top-1 repair as LOCUS' label-free proposal on every dataset: LOCUS is a verifier first, and its repair head, trained only on synthetic corruptions, is a useful by-product rather than a state-of-the-art corrector. Scoring ∼17 hypotheses with two recognisers and two renderings runs at about 18 words/s on one GPU, slower than a single PARSeq pass and about 3× slower than a batched 7B judge, although its frozen models are 23× smaller (Table 8).

6. Analysis: What Drives the Gains?

6.1 Statistical significance at the document level

Words cropped from one page share scanner, font and annotator, so a word-level bootstrap overstates confidence. For FUNSD and CORD we recovered the source document of every crop (by replaying the seeded crop selection, verified against the released manifest with zero

Table 7. Human-review workload: percentage of all words that must be reviewed so that the automatically accepted words reach the target precision ($k = 25$; lower is better).

Dataset	Method	90%	95%	98%
FUNSD	LOCUS	**30.1**	37.2	61.1
	PARSeq re-OCR (graded)	30.4	**35.4**	**56.2**
	TrOCR-base re-OCR	69.8	100.0	100.0
	Qwen2.5-VL-7B judge	57.1	85.3	100.0
	Engine confidence	47.9	65.5	99.9
CORD	LOCUS	**33.6**	**37.4**	**39.6**
	PARSeq re-OCR (graded)	34.8	38.6	40.6
	TrOCR-base re-OCR	50.8	55.3	57.3
	Qwen2.5-VL-7B judge	36.7	64.9	95.5
	Engine confidence	49.4	59.2	76.0
SROIE	LOCUS	**41.2**	**44.7**	**54.9**
	PARSeq re-OCR (graded)	66.9	69.5	71.7
	TrOCR-base re-OCR	41.9	45.0	94.6
	Qwen2.5-VL-7B judge	53.0	76.2	92.3
	Engine confidence	67.9	90.9	97.0
IAM	LOCUS	**44.3**	**51.4**	66.9
	PARSeq re-OCR (graded)	51.3	54.6	**62.5**
	TrOCR-base re-OCR	99.9	99.9	99.9
	Qwen2.5-VL-7B judge	86.9	94.8	96.8
	Engine confidence	63.9	90.9	99.8
IIIT5K	LOCUS	21.3	25.6	28.1
	PARSeq re-OCR (graded)	**21.1**	**25.3**	**27.7**
	TrOCR-base re-OCR	24.2	28.5	31.0
	Qwen2.5-VL-7B judge	23.0	28.4	34.2
	Engine confidence	26.1	36.9	47.8

Table 8. Repair and cost. Left: share of real errors whose truth lies in $G(h)$, and top-1 exact repair accuracy (%) of substituting TrOCR's reading, PARSeq's reading, or LOCUS' label-free proposal. Right: measured throughput on 256 FUNSD words on one RTX A6000.

Dataset	errors	in $G(h)$	TrOCR	PARSeq	LOCUS
FUNSD	1474	82.8	31.3	**72.4**	67.7
CORD	941	94.9	70.4	**89.5**	81.4
SROIE	1880	92.6	**89.5**	44.1	**89.5**
IAM	1857	55.4	14.4	**46.5**	29.0
IIIT5K	869	97.0	86.8	**95.3**	93.6

Component	parameters	words/s
LOCUS evidence extraction (TrOCR-base + PARSeq, both renderings)	358M (frozen)	17.9
LOCUS GNN (trainable)	0.05M	–
PARSeq re-recognition	24M	372.3
Florence-2-base re-recognition	0.23B	14.9
Tesseract 5 (CPU, 1 thread)	–	9.3
Qwen2.5-VL-7B judge	8.3B	53.7

mismatches) and resample whole *documents* (50 test forms, 100 test receipts). IIIT5K has one word per image, so word-level resampling is exact; the public SROIE and IAM releases carry no document identifiers, so their intervals are word-level and may be optimistic (Table 9). Against the strongest baseline, LOCUS is significantly better on IAM in both metrics (ΔAUROC $+0.055$, ΔF1 $+0.026$, $p < 0.001$), better in AUROC on CORD at the document level ($+0.007$, $p = 0.023$; macro-F1 $+0.005$, not significant) and on SROIE ($+0.014$, $p < 0.001$, while TrOCR re-recognition is slightly better in macro-F1, -0.006, $p = 0.002$); it is statistically tied with PARSeq on FUNSD at the document level ($p = 0.25$ and $p = 0.12$) and significantly worse than PARSeq on IIIT5K (-0.004 AUROC, -0.021 macro-F1). It is significantly better than the VLM judge and than TrOCR re-recognition on every dataset except SROIE, where the latter is in-domain.

6.2 How much does LOCUS depend on its recognisers?

Table 10 lists each verifier's word accuracy and LOCUS' gain over the better re-recognition baseline; Table 11 splits every test set by *which* verifier reads the word correctly. Three regularities hold on all five datasets. (i) When both verifiers read the word correctly every method is nearly perfect. (ii) When only one verifier is right, re-recognition built on the

Table 9. Paired bootstrap (2,000 resamples) of LOCUS ($k = 25$) against the strongest baselines: difference in AUROC and in macro-F1, 95% interval and two-sided p. Resampling unit: documents for FUNSD and CORD, words otherwise.

Dataset	unit	Baseline	ΔAUROC	95% CI	p	ΔF1	95% CI	p
FUNSD	50 docs	PARSeq re-OCR (graded)	-0.002	[-0.007, +0.002]	0.252	-0.004	[-0.008, +0.001]	0.121
FUNSD	50 docs	TrOCR-base re-OCR	+0.299	[+0.266, +0.330]	<0.001	+0.338	[+0.287, +0.389]	<0.001
FUNSD	50 docs	Qwen2.5-VL-7B judge	+0.128	[+0.111, +0.146]	<0.001	+0.183	[+0.168, +0.198]	<0.001
CORD	100 docs	PARSeq re-OCR (graded)	+0.007	[+0.001, +0.015]	0.023	+0.005	[-0.001, +0.012]	0.122
CORD	100 docs	TrOCR-base re-OCR	+0.152	[+0.122, +0.184]	<0.001	+0.167	[+0.128, +0.206]	<0.001
CORD	100 docs	Qwen2.5-VL-7B judge	+0.049	[+0.036, +0.063]	<0.001	+0.088	[+0.071, +0.105]	<0.001
SROIE	4000 words	PARSeq re-OCR (graded)	+0.232	[+0.217, +0.246]	<0.001	+0.232	[+0.218, +0.246]	<0.001
SROIE	4000 words	TrOCR-base re-OCR	+0.014	[+0.010, +0.017]	<0.001	-0.006	[-0.010, -0.002]	0.002
SROIE	4000 words	Qwen2.5-VL-7B judge	+0.051	[+0.044, +0.058]	<0.001	+0.107	[+0.097, +0.118]	<0.001
IAM	4000 words	PARSeq re-OCR (graded)	+0.055	[+0.047, +0.063]	<0.001	+0.026	[+0.015, +0.037]	<0.001
IAM	4000 words	TrOCR-base re-OCR	+0.203	[+0.192, +0.214]	<0.001	+0.148	[+0.134, +0.161]	<0.001
IAM	4000 words	Qwen2.5-VL-7B judge	+0.181	[+0.169, +0.194]	<0.001	+0.206	[+0.191, +0.222]	<0.001
IIIT5K	2999 words	PARSeq re-OCR (graded)	-0.004	[-0.007, -0.001]	0.005	-0.021	[-0.028, -0.016]	<0.001
IIIT5K	2999 words	TrOCR-base re-OCR	+0.025	[+0.019, +0.031]	<0.001	+0.019	[+0.010, +0.027]	<0.001
IIIT5K	2999 words	Qwen2.5-VL-7B judge	+0.017	[+0.012, +0.023]	<0.001	+0.083	[+0.072, +0.094]	<0.001

Table 10. Recogniser dependence. Word accuracy (%) of each verifier's reading, the better re-recognition baseline, and LOCUS ($k = 25$) against it.

	verifier accuracy		best	macro-F1			AUROC		
Dataset	TrOCR	PARSeq	baseline	base	LOCUS	Δ	base	LOCUS	Δ
FUNSD	38.0	88.9	PARSeq	0.946	0.943	-0.004	0.976	0.974	-0.002
CORD	69.8	94.0	PARSeq	0.973	0.978	+0.005	0.986	0.993	+0.007
SROIE	94.3	49.5	TrOCR-base	0.979	0.973	-0.006	0.978	0.992	+0.014
IAM	42.4	64.7	PARSeq	0.879	0.906	+0.027	0.914	0.969	+0.055
IIIT5K	92.7	98.5	PARSeq	0.993	0.971	-0.021	0.995	0.991	-0.004

other verifier fails badly, while LOCUS stays accurate because it weighs the evidence of both (IAM: 89.5% vs. 25.7% for PARSeq agreement on words only TrOCR reads, 89.0% vs. 43.6% for TrOCR agreement on words only PARSeq reads). (iii) When neither is right, all methods degrade. LOCUS' gain is therefore governed by how often the verifiers *disagree in correctness*: it is largest on IAM, where each verifier alone is right on only 42–65% of words, and vanishes on IIIT5K, where PARSeq is right on 98.5%. LOCUS is best understood as a learned, label-efficient combination of complementary recognisers through their counterfactual evidence, not as a substitute for a strong in-domain recogniser.

6.3 Why counterfactual evidence matters more than message passing

Removing message passing does not hurt (Table 5). To understand why, Table 12 runs five controlled variants, each pretrained label-free with the same change. Three observations explain the null effect of message passing. First, removing message passing changes neither metric by more than 0.009 on any dataset (A vs. B). Second, the *explicit* comparison of each rival with the claim does matter: removing the claim-relative features costs 0.010 to 0.023 macro-F1 on CORD, SROIE and IIIT5K (A vs. C), and adding message passing back does not recover it (C vs. D). The network is not learning the comparison from the graph; it uses the comparison

Table 11. Decision accuracy (%) stratified by which verifier reads the word correctly ($k = 25$; err.: share of incorrect claims in the stratum; strata with fewer than 20 words omitted).

Dataset	stratum	words	err. (%)	LOCUS	PARSeq agree.	TrOCR agree.
FUNSD	both right	1475	29	99.5	**100.0**	**100.0**
	only TrOCR right	44	80	70.5	47.7	**100.0**
	only PARSeq right	2080	31	98.7	**100.0**	29.0
	neither right	401	93	60.1	56.9	**76.6**
CORD	both right	1592	39	**100.0**	**100.0**	**100.0**
	only TrOCR right	52	71	92.3	61.5	**100.0**
	only PARSeq right	622	35	97.9	**100.0**	34.2
	neither right	90	69	**64.4**	54.4	61.1
SROIE	both right	1946	41	99.5	**100.0**	**100.0**
	only TrOCR right	1827	48	98.1	47.0	**100.0**
	only PARSeq right	33	88	63.6	**100.0**	51.5
	neither right	194	87	73.2	**82.5**	64.4
IAM	both right	1400	13	99.3	**100.0**	**100.0**
	only TrOCR right	296	27	89.5	25.7	**100.0**
	only PARSeq right	1187	57	89.0	**100.0**	43.6
	neither right	1117	82	**81.9**	76.5	73.6
IIIT5K	both right	2764	27	98.9	**100.0**	**100.0**
	only TrOCR right	15	–	–	–	
	only PARSeq right	190	45	86.3	**100.0**	42.6
	neither right	30	97	50.0	53.3	**60.0**

Table 12. Message passing versus counterfactual evidence ($k = 25$, macro-F1 / AUROC, ensembles of five seeds; best in bold, second best underlined). Claim-relative features: the differences $\phi(v) - \phi(h)$ between each rival and the claim and the claim's agreement with the verifiers' readings.

Variant	FUNSD	CORD	SROIE	IAM	IIIT5K
A full	.943 / .974	**.979** / .993	**.973** / **.992**	.901 / **.967**	.970 / **.991**
B no message passing	**.945** / **.975**	.978 / **.994**	.970 / **.992**	.895 / .966	**.979** / **.991**
C no claim-relative features	.936 / .972	.969 / .993	.950 / .988	.898 / .964	.954 / .988
D no claim-relative features, no MP	.943 / .974	.967 / **.994**	.956 / .990	.882 / .960	.960 / .987
E claim node only (no rivals)	.944 / .961	.978 / .993	.972 / **.992**	**.914** / **.967**	.962 / .990

when it is computed for it. Third, the graph is a depth-one star whose readout already pools over every rival, so a message-passing layer can only re-express information the readout sees anyway. Removing the rivals entirely (E) costs ranking quality on FUNSD (AUROC .974 to .961) but little elsewhere, because the claim node keeps its own evidence and its agreement with both verifiers' readings, which summarise what the strongest rivals would say. In short, what helps is counterfactual *evidence*: scoring the claim against alternatives under prior-cancelled likelihoods and making the comparison explicit; the graph is a convenient container for it, not the source of the gain.

6.4 Sensitivity to the rendering

The ratio depends on a rendering $R(v)$ whose font never matches the document. We re-extracted all evidence with four other fonts (including a handwriting-like one) and two other glyph sizes on fixed subsets of FUNSD, CORD and IAM (1,000 training and 1,000 test words each) and re-evaluated the claim's ratio alone, the training-free ρ-margin rule, and LOCUS trained on the

Table 13. Rendering sensitivity: AUROC of the claim's ratio $\rho_{V_1} + \rho_{V_2}$ alone, AUROC of the training-free ρ-margin rule, and macro-F1 of LOCUS trained on 1,000 labelled words (3 seeds), for each rendering.

	FUNSD			CORD			IAM		
Rendering	$\rho(h)$	rule	LOCUS	$\rho(h)$	rule	LOCUS	$\rho(h)$	rule	LOCUS
Liberation Sans (default)	.903	.923	.937	.981	**.996**	.972	.921	.921	.921
Liberation Serif	.906	.928	.938	.984	.995	.972	.918	**.944**	.920
Liberation Mono	.907	**.940**	.938	**.990**	.994	**.977**	.915	.931	**.923**
DejaVu Sans	**.912**	.920	.937	.984	.995	.976	.922	.923	.919
Comic Neue (handwriting-like)	.907	.928	**.939**	.989	.993	.976	.917	.937	.921
Liberation Sans, glyphs 0.6x	.903	.924	**.939**	.983	**.996**	.976	.915	.919	**.923**
Liberation Sans, glyphs 0.9x	.901	.924	.937	.980	.995	.973	**.923**	.923	.921

Table 14. AUROC on all errors / on genuine character errors only ($k = 25$). Removed: share of incorrect claims that differ from the ground truth only in case or punctuation.

Dataset	removed	LOCUS	PARSeq graded	TrOCR re-OCR	Qwen2.5-VL-7B	Engine conf.
FUNSD	51%	0.974 / 0.976	**0.976 / 0.979**	0.675 / 0.691	0.846 / 0.928	0.861 / 0.886
CORD	19%	**0.993 / 0.996**	0.986 / 0.987	0.841 / 0.845	0.945 / 0.989	0.892 / 0.907
SROIE	23%	**0.992 / 0.999**	0.760 / 0.756	0.978 / 0.991	0.941 / 0.982	0.859 / 0.886
IAM	18%	**0.969 / 0.980**	0.914 / 0.908	0.766 / 0.802	0.788 / 0.818	0.878 / 0.916
IIIT5K	0%	0.991 / 0.991	**0.995 / 0.995**	0.966 / 0.966	0.974 / 0.974	0.938 / 0.938

subset (Table 13). The trained model is insensitive to the rendering: across seven renderings its macro-F1 varies by at most 0.005 on every dataset (FUNSD .937–.939, CORD .972–.977, IAM .919–.923), and the AUROC of the claim's ratio alone by at most 0.011. The training-free rule is more sensitive (AUROC .920–.940 on FUNSD, .919–.944 on IAM); it is best with a serif or monospaced font rather than our default sans-serif, which we did not tune, and a handwriting-like font does not help on handwriting. A mismatched rendering is thus absorbed by learning; only the label-free rule would benefit from choosing the font.

6.5 Robustness to annotation and cropping artefacts

Table 14 re-scores every method after removing incorrect claims that differ from the ground truth only in case or punctuation, which the audit (Section 7) shows are often annotation conventions, label noise or fragments of neighbouring glyphs. The removed share is large on FUNSD (51% of errors) and moderate on CORD, SROIE and IAM (18–23%); IIIT5K's canonical protocol already excludes such differences. LOCUS' AUROC changes by at most 0.011 (it rises slightly or is unchanged), and its ordering relative to the baselines is unchanged except that the VLM judge, whose AUROC rises most (FUNSD .846 to .928, SROIE .941 to .982), is visibly penalised by these artefacts. The conclusions of Sec. 5.1 therefore do not rest on annotation conventions.

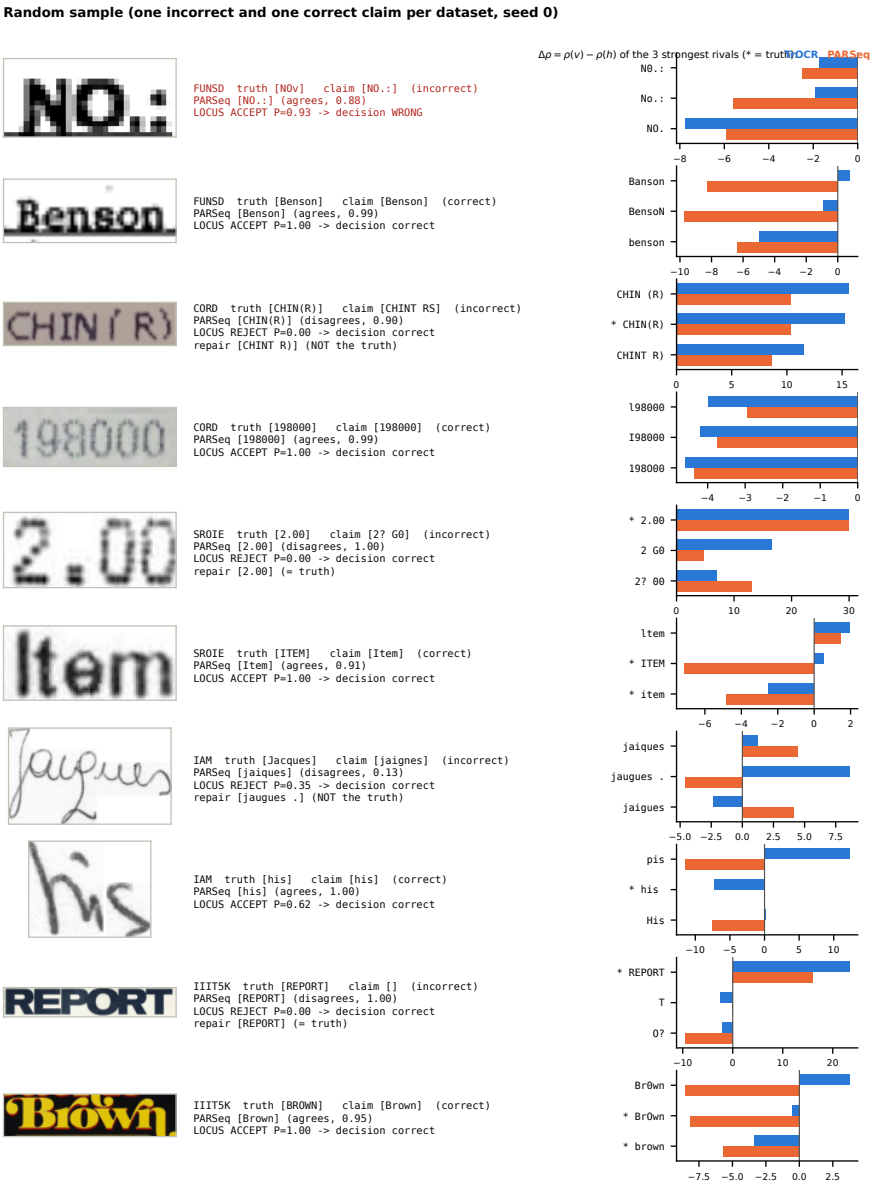


Figure 8. Random sample (seed 0): one incorrect and one correct claim per dataset, drawn without regard to LOCUS' output; wrong decisions in red. Annotation errors also appear (e.g. FUNSD ground truth "NOv" for a crop reading "NO.:"; Table 16).

7. Qualitative Analysis and Manual Audit

Every row below is drawn by a seeded random choice inside a rule-defined stratum, and LOCUS' *failures* are shown as often as its successes. Each row shows the crop, the ground truth, the claim, PARSeq's reading, LOCUS' probability and whether its decision is right (red: wrong), the repair proposal for rejected incorrect claims, and the evidence of the three strongest rivals relative to the claim, $\Delta\rho_V(v) = \rho_V(v) - \rho_V(h)$, for TrOCR (blue) and PARSeq (orange); a positive bar means that verifier finds the rival a better explanation of the image than the claim, and $*$ marks the ground truth.

Regimes. Table 15 counts the two regimes that matter on the full test sets. Correct claims that

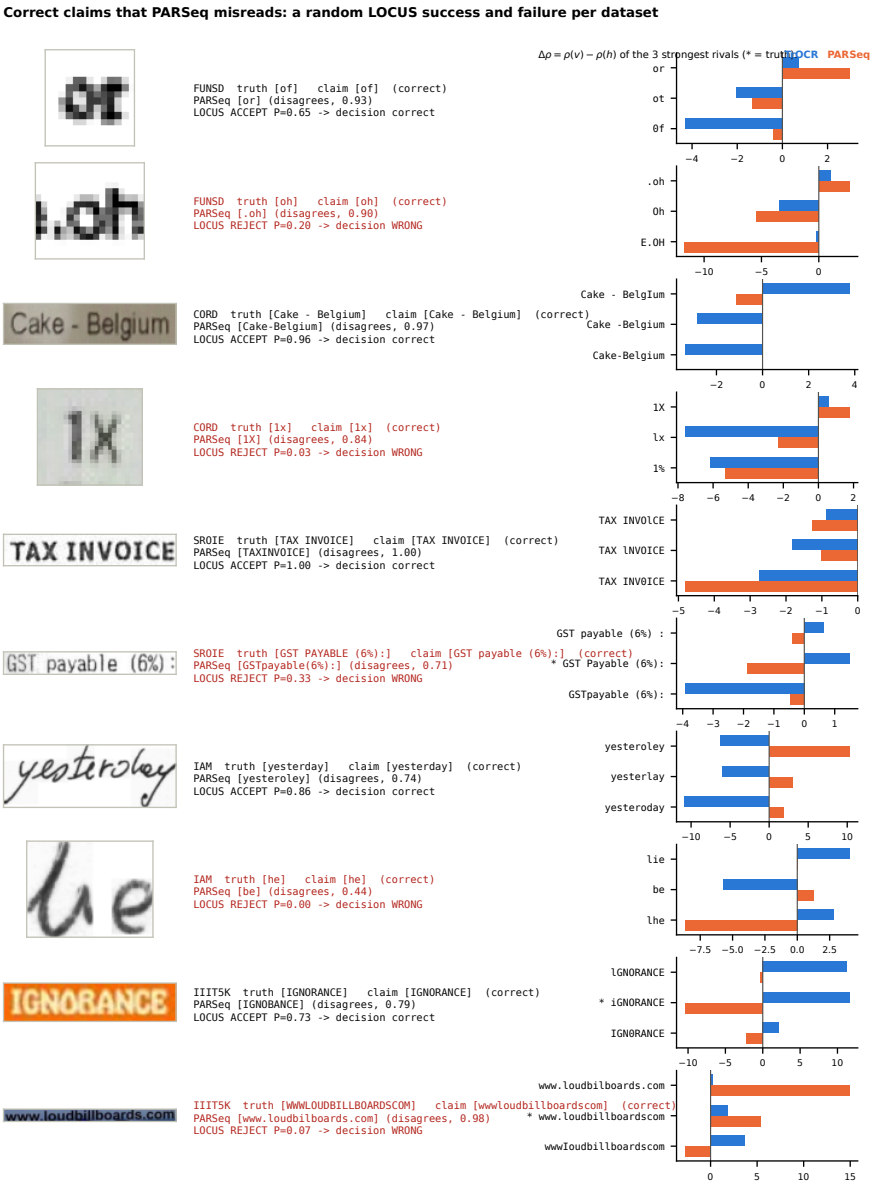


Figure 9. Correct claims that PARSeq misreads (re-recognition would reject them): one random LOCUS success and one random failure per dataset. Comparing the claim with its rivals under two verifiers helps most in this regime: a rival that one verifier prefers is typically rejected by the other.

PARSeq misreads are frequent on every dataset, and LOCUS accepts most of them (Figure 9); this is the regime in which comparing hypotheses under two verifiers pays off. Character errors that PARSeq reproduces are rarer and LOCUS rejects only a small fraction of them (Supplementary Figure S5).

Manual audit. To understand the errors behind these numbers, we inspected two random samples of up to 20 test words per dataset: (i) words on which LOCUS decides wrongly and (ii) incorrect claims that PARSeq agrees with. Each crop was labelled *genuine* (the ground truth matches the visible text), *label noise*, *crop problem* (text clipped or a neighbouring glyph inside the box) or *ambiguous* (Table 16). Two conclusions follow. First, fewer than half of LOCUS' apparent mistakes are genuine verification errors on a correct label (42 of 100 sampled): 25 are

Table 15. Regimes on the full test sets ($k = 25$). Artefacts: incorrect claims that differ from the ground truth only in case or punctuation. Right: errors PARSeq agrees with, restricted to genuine character errors.

	all errors		correct claims PARSeq misreads		character errors PARSeq agrees with	
Dataset	count	artefacts	count	LOCUS accepts	count	LOCUS rejects
FUNSD	1474	751 (51%)	38	14 (37%)	47	3 (6%)
CORD	941	181 (19%)	43	28 (65%)	9	1 (11%)
SROIE	1880	438 (23%)	971	953 (98%)	7	4 (57%)
IAM	1857	326 (18%)	419	312 (74%)	53	14 (26%)
IIIT5K	869	0 (0%)	4	3 (75%)	14	0 (0%)

Table 16. Manual audit of random test words (20 per cell, fewer where the stratum is smaller). G: genuine, N: label noise, C: crop problem, A: ambiguous, P: claim correct only under the dataset's normalisation protocol although its exact appearance differs from the image (IIIT5K only).

	LOCUS decision wrong					errors PARSeq agrees with			
Dataset	G	N	C	A	P	G	N	C	A
FUNSD	8	6	4	2	0	0	16	3	1
CORD	10	3	4	3	0	1	8	4	5
SROIE	7	12	1	0	0	5	10	1	4
IAM	16	1	0	3	0	4	1	11	4
IIIT5K	1	3	0	0	16	0	11	2	1

label noise, 9 crop problems and 8 ambiguous, and on IIIT5K 16 of 20 are claims that count as correct only after the case-insensitive, punctuation-free normalisation of the protocol (*e.g.* `HoLLywooD` for an upper-case sign), which LOCUS, judging the exact appearance of the string, rejects; this protocol mismatch, rather than a visual error, explains most of LOCUS' gap to PARSeq agreement (which compares normalised strings) on IIIT5K. Second, the stratum "errors that PARSeq agrees with" consists mostly of cases where the claim matches what is visible: only 10 of 92 sampled words are genuine errors, against 46 label-noise and 21 crop problems, so low rejection rates there are not a weakness of either verifier. The genuine failures that remain fall into two identifiable modes: rejecting correct short or case-ambiguous claims when PARSeq reads another case (CORD: `Rp`, `x`, `Pop`), and accepting capitalisation errors in handwriting (IAM: `pinkie`, `market`, `church`). The audit was performed by a single annotator; the annotated sheets are released with the code.

8. Discussion and Limitations

When does comparing hypotheses help? LOCUS helps when its two recognisers *disagree in correctness*: on words that only one verifier reads correctly, re-recognition with the other verifier is right only 26–62% of the time, while LOCUS is right 64–99% of the time on every dataset (Table 11). It therefore gains most where recognisers are individually fallible and complementary (handwriting, camera receipts) and least where one in-domain recogniser is nearly always right (PARSeq on IIIT5K, TrOCR-printed on SROIE), and it does not require knowing in advance

which recogniser to trust: the best re-recognition baseline changes from dataset to dataset, whereas LOCUS is close to the best on all of them.

Limitations. All design decisions were made on FUNSD and CORD; SROIE, IAM and IIIT5K are held out. (i) Repair is bounded by whether the truth lies in $G(h)$, which holds for only about half of the handwriting errors (Table 2); verification degrades more gracefully because a claim can lose to a rival that is not the truth. (ii) Pretraining relies on engine agreement, whose precision ranges from 84% to 99% (Supplementary Table S1); it helps on four datasets but not on IAM, where agreement is least precise. (iii) TrOCR's printed checkpoints were fine-tuned on SROIE and PARSeq was developed on scene-text benchmarks such as IIIT5K; the re-recognition baselines use the same recognisers, so the comparison is on equal footing, but absolute numbers there should be read with this in mind. (iv) Message passing, the local-optimality term and render normalisation are individually redundant in the full model, and simple learners over the same evidence are almost as good: the counterfactual evidence, not the network, is the main contribution. (v) LOCUS judges whether the exact string, including case and punctuation, explains the image; under protocols that normalise both away (IIIT5K) a claim such as `HoLLywooD` counts as correct but looks wrong. (vi) All datasets are Latin-script. (vii) Evidence extraction runs at about 18 words/s on one GPU; pruning the neighbourhood or sharing encoder passes across hypotheses are straightforward speed-ups. Deployments should monitor accepted precision on audited samples.

9. Conclusion

We reframed OCR verification as asking whether a claim is the locally optimal explanation of its image among OCR-plausible counterfactual alternatives, and made the comparison visual with a render-normalised likelihood ratio under two frozen recognisers. LOCUS, a relation-aware graph network over counterfactual string graphs, is pretrained without human labels and adapts from a handful of labelled words. On a new benchmark of real OCR errors over five public datasets it outperforms all general-purpose verifiers in AUROC on every dataset and, against re-recognition with state-of-the-art recognisers, is the only verifier close to the best everywhere. Our analyses locate the benefit in the counterfactual evidence and label-free pretraining rather than in deep message passing. Future work will extend the confusion model and verifiers to non-Latin scripts and make the evidence aware of each dataset's comparison protocol.

Declarations

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Ethical statement. This study uses only publicly available datasets (FUNSD, CORD, SROIE,

IAM and IIIT5K) under their respective licences and publicly released pretrained models. It involves no human participants, animals or personal data beyond what the public datasets contain, and therefore required no ethical approval.

Conflicts of interest / competing interests. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability. The benchmark manifests, all code, raw result files and the annotated audit sheets are available at an anonymised repository (link withheld for double-blind review).

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